

Lab 5 – 20.05.2026

During the class, we will follow a course that has been proposed by Google:

<https://developer.android.com/courses/android-basics-compose/course>

Tasks for the fifth lab:

1. Unit 5: Connect to the Internet

<https://developer.android.com/courses/android-basics-compose/unit-5>

2. Pathway 1: Get Data from the Internet

<https://developer.android.com/courses/pathways/android-basics-compose-unit-5-pathway-1>

Implement coroutines to perform tasks concurrently without blocking the app, and learn about HTTP and REST to get data from the internet.

- a. Introduction to Coroutines in Kotlin Playground
<https://developer.android.com/codelabs/basic-android-kotlin-compose-coroutines-kotlin-playground>
- b. Introduction to Coroutines in Android Studio
<https://developer.android.com/codelabs/basic-android-kotlin-compose-coroutines-android-studio>
- c. Get data from the internet
<https://developer.android.com/codelabs/basic-android-kotlin-compose-getting-data-internet>

3. Pathway 2: Load and Display Images from the Internet

<https://developer.android.com/courses/pathways/android-basics-compose-unit-5-pathway-2>

Apply architecture best practices to the app and use Coil to download and display images.

- a. Add repository and Manual Dependency Injection
<https://developer.android.com/codelabs/basic-android-kotlin-compose-add-repository>
- b. Load and display images from the internet
<https://developer.android.com/codelabs/basic-android-kotlin-compose-load-images>
- c. Practice: Build amphibians app (**mandatory**)
<https://developer.android.com/codelabs/basic-android-kotlin-compose-practice-amphibians-app>

- d. Project: Create a Bookshelf app (**mandatory**)
<https://developer.android.com/codelabs/basic-android-kotlin-compose-bookshelf>

As a result of the work please take the Quiz at the end of each Pathway.

Submission requirements:

- Use the **private** repository named MAD created during Lab 2.
- Inside the repository maintain the following folder structure:

```
MAD/  
  UNIT5/  
    AmphibiansApp      (Practice: Build amphibians app)  
    BookshelfApp/      (Project: Create a Bookshelf app)
```

- The instructor should already be a collaborator (rmaciaszczyk@zut.edu.pl). If not, add them using that email address.
- Submit the repository link as a response to this assignment in **Microsoft Teams**.